

lab_3_exercise

November 23, 2024

1 Objectives

The objective of this lab is to practice essential data analysis techniques using Pandas. It focuses on performing fundamental operations with Pandas, including creating, manipulating, and analyzing data. Through hands-on tasks, you will learn how to:

1. **Create DataFrames:** Understand how to construct DataFrames from dictionaries and other sources.
2. **Read and Write Data:** Work with various file types like CSV and Excel files to load and save datasets.
3. **Perform Basic Statistical Analysis:** Calculate descriptive statistics such as mean, standard deviation, and transformations of data columns.
4. **Data Cleaning and Transformation:** Handle missing values, apply mathematical operations, and reshape data.
5. **Group and Aggregate Data:** Use grouping and aggregation techniques to summarize data effectively.

The overall goal is to build confidence in using Pandas for real-world data analysis tasks and lay a foundation for more advanced analytics workflows.

1.1 Question 1: Recreate the DataFrame Using from_dict

```
data = {'year': [2010, 2011, 2012, 2010, 2011, 2012, 2010, 2011, 2012],
'team': ['FCBarcelona', 'FCBarcelona', 'FCBarcelona', 'RMadrid', 'RMadrid', 'RMadrid', 'Valencia', 'Valencia', 'Valencia'],
'wins': [30, 28, 32, 29, 32, 26, 21, 17, 19],
'draws': [6, 7, 4, 5, 4, 7, 8, 10, 8],
'losses': [2, 3, 2, 4, 2, 5, 9, 11, 11]}
football = pd.DataFrame(
data, columns=['year', 'team', 'wins', 'draws', 'losses'])
football
```

Task: Recreate the DataFrame above using the `from_dict` method.

- The `from_dict` method creates a DataFrame from a dictionary. Each key becomes a column, and the corresponding values become the data.

```
[4]: import pandas as pd
```

```
# Recreate the DataFrame using from_dict
data = {
    'year': [2010, 2011, 2012, 2010, 2011, 2012, 2010, 2011, 2012],
    'team': ['FCBarcelona', 'FCBarcelona', 'FCBarcelona',
            'RMadrid', 'RMadrid', 'RMadrid',
            'ValenciaCF', 'ValenciaCF', 'ValenciaCF'],
    'wins': [30, 28, 32, 29, 32, 26, 21, 17, 19],
    'draws': [6, 7, 4, 5, 4, 7, 8, 10, 8],
    'losses': [2, 3, 2, 4, 2, 5, 9, 11, 11]
}

football = pd.DataFrame.from_dict(data)
print(football)
```

	year	team	wins	draws	losses
0	2010	FCBarcelona	30	6	2
1	2011	FCBarcelona	28	7	3
2	2012	FCBarcelona	32	4	2
3	2010	RMadrid	29	5	4
4	2011	RMadrid	32	4	2
5	2012	RMadrid	26	7	5
6	2010	ValenciaCF	21	8	9
7	2011	ValenciaCF	17	10	11
8	2012	ValenciaCF	19	8	11

1.2 Question 2: Read a Non-CSV File

Task: Read a non-CSV file (e.g., Excel) into a DataFrame.

- The `pd.read_excel()` function is used to read Excel files.
- Use the `head()` method to display the first few rows.

```
[5]: # Example of reading an Excel file
excel_file_path = 'heads_of_state.xlsx'
q_2 = pd.read_excel(excel_file_path)
print(q_2.head())
```

	Unnamed: 0		Name	Took office	Left office	\
0	1		Dr. Nanmdi Azikiwe	1960-11-16	1966-01-16	
1	2	Major General	Johnson Aguiyi-Ironsi	1966-01-16	1966-07-29	
2	3		General Yakubu Gowon	1966-08-01	1975-07-29	
3	4		General Murtala Mohammed	1975-07-29	1976-02-13	
4	5		General Olusegun Obasanjo	1976-02-13	1979-10-01	

	Military/Civilian	Age	Took Office	Died in Office?	Days in Office
0	Civilian		56	No	1887.0
1	Military		42	Yes	194.0
2	Military		32	No	3284.0
3	Military		37	Yes	199.0

1.2.1 Question 3: Calculate the Mean and Standard Deviation of the Value Column

Task: Compute the average and standard deviation of the Value column in the edu DataFrame.

- The `mean()` and `std()` methods compute the mean and standard deviation, excluding NaN values by default.

```
[6]: edu = pd.read_csv('education_analysis.csv', na_values=':', usecols=['TIME', 'GEO', 'Value'])

# Calculate mean and standard deviation
mean_value = edu['Value'].mean()
std_value = edu['Value'].std()

print(f"Mean: {mean_value}")
print(f"Standard Deviation: {std_value}")
```

Mean: 5.20398891966759

Standard Deviation: 1.0216944748195935

1.2.2 Question 4: Calculate the Ceil of the Scalar Square Root of Value

Task: Compute the square root of the Value column, take its ceil, and display the result.

- Use `apply()` to apply a function to each value in the column.
- Use `np.sqrt()` to calculate the square root and `np.ceil()` to round up.

```
[7]: import numpy as np

# Compute the square root and ceil
edu['sqrt_ceil'] = edu['Value'].apply(lambda x: np.ceil(np.sqrt(x)) if pd.
    notnull(x) else np.nan)
print(edu[['Value', 'sqrt_ceil']].head())
```

	Value	sqrt_ceil
0	NaN	NaN
1	NaN	NaN
2	5.00	3.0
3	5.03	3.0
4	4.95	3.0

1.2.3 Question 5: Calculate the Mean of Value for Each Country Per Year

Task: Group the data by GEO and TIME to calculate the mean Value.

- The `groupby()` method groups the data by GEO (country) and TIME (year).
- The `mean()` function calculates the mean for each group.

```
[8]: # Group by GEO and TIME, then calculate mean
mean_per_year = edu.groupby(['GEO', 'TIME'])['Value'].mean()
print(mean_per_year)
```

GEO	TIME	
Austria	2000	5.66
	2001	5.74
	2002	5.68
	2003	5.53
	2004	5.48
...		
Spain	2007	4.34
	2008	4.62
	2009	5.02
	2010	4.98
	2011	4.82

Name: Value, Length: 384, dtype: float64

2 Summary

In this lab, I explored the fundamental functionalities of Pandas for data analysis and manipulation. Starting with the creation of DataFrames using the `from_dict` method, I learned how to structure data in tabular format. I also practiced reading data from external files like CSV and Excel, with a focus on handling missing values by specifying custom markers (`na_values`).

- The solutions illustrate key Pandas functionalities like creating DataFrames, reading data, filtering, grouping, and aggregation.
- Each section in the notebook includes comments and explanations to demonstrate the thought process.

```
[ ]:
```